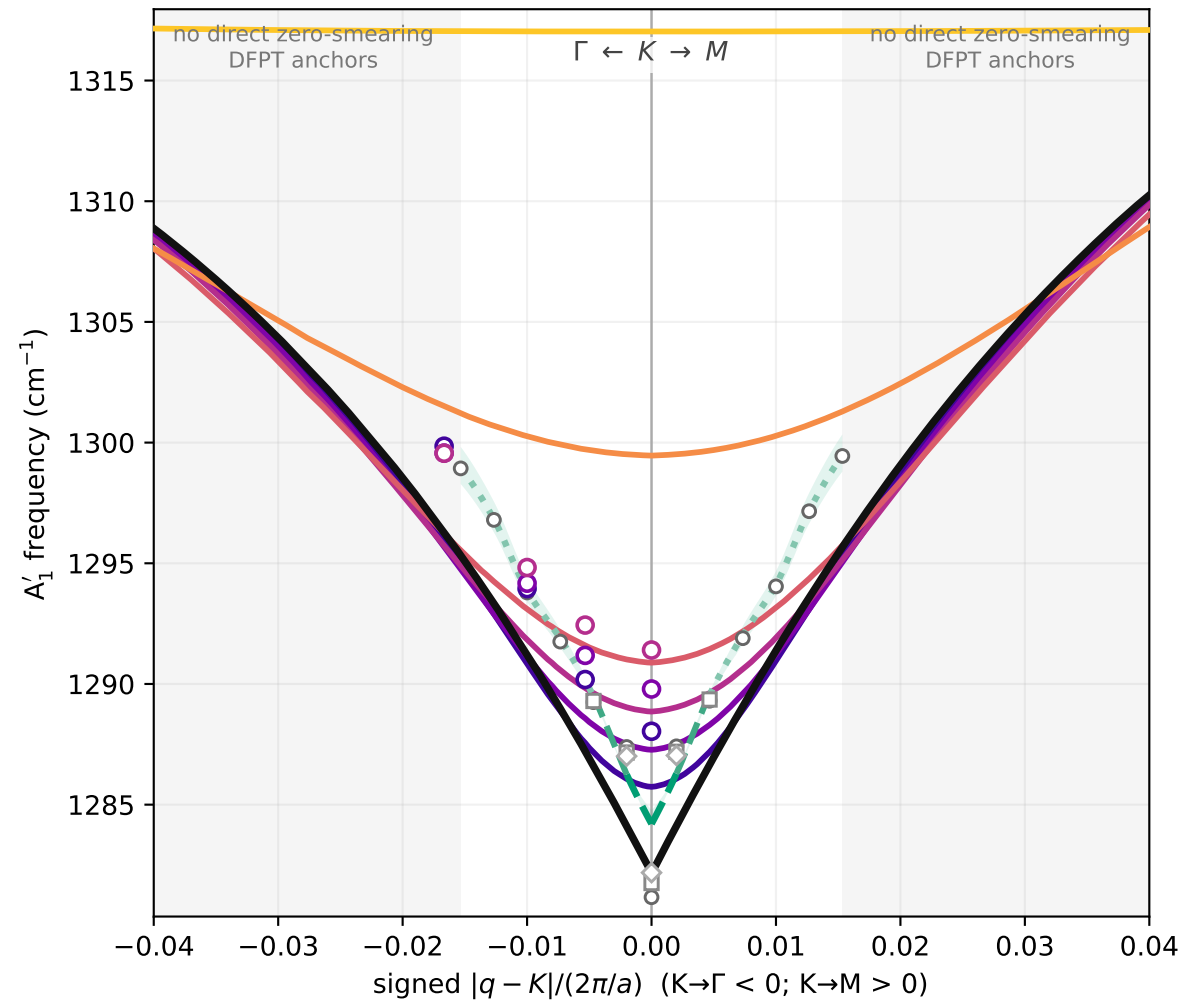
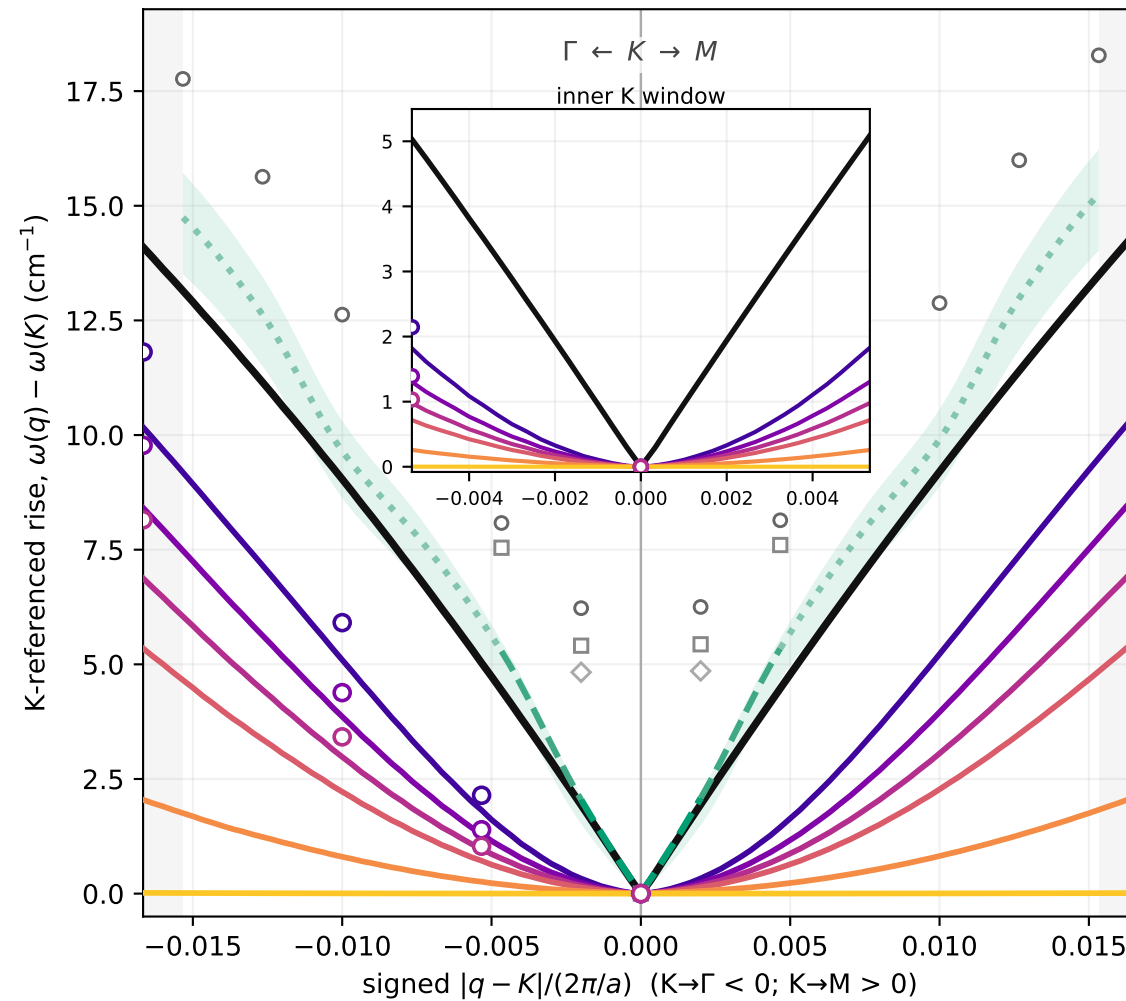


Graphene K-point A'_1 : electronic smearing at one fixed lattice condition

(a) Absolute frequency: extended K region



(b) Cusp shape: each series referenced to its own K



MLIP + full EPC LR
smearing/degauss (Ry)

- 0
- 0.0019001
- 0.0028501
- 0.0038002
- 0.0050000
- 0.0100000
- 0.0200000

- zero-smearing DFPT $N \rightarrow \infty$ guide
- zero-smearing guide envelope
- raw zero-smearing DFPT (k192/240/288)
- direct finite-smearing DFPT ($K \rightarrow \Gamma$ only)
- outside direct zero-smearing DFPT support

Lattice condition held fixed:
same static MLIP $D_{\text{SR}}(q)$ for all 7 curves;
no 300/450/600 K lattice/TDEP backgrounds.

One common absolute EPC reference constant;
no per-smearing K re-anchoring.
0.020 Ry is the frozen response reference.
Finite-smearing markers: direct static-lattice DFPT.